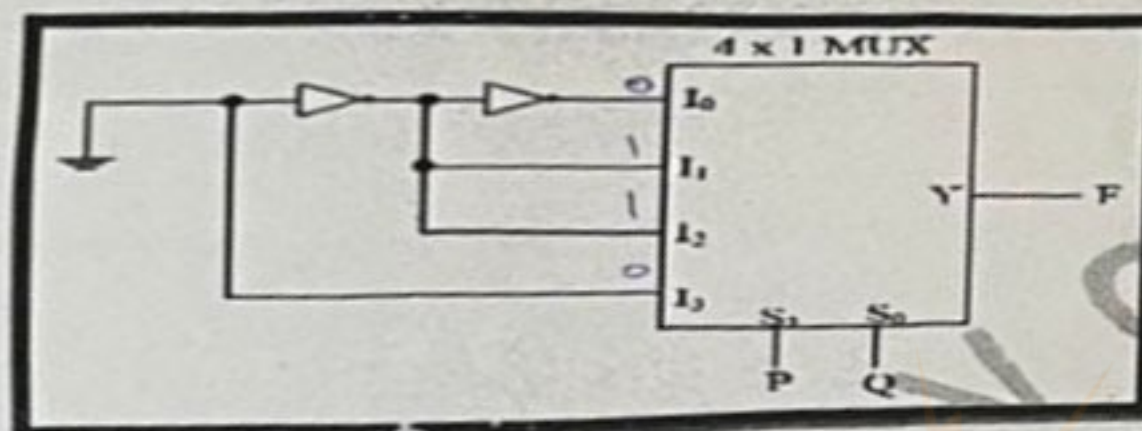


Q1 A) Consider the Combinational circuit below, Determine:

- i. The logic function implemented (ground implies a logic "0")

[4 marks]



Solution

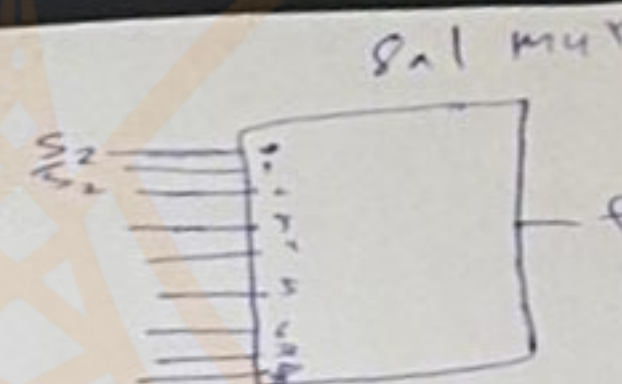
$$F = S_1 + S_0$$

[4 marks]

- ii. Implement the circuit using an 8x1 Mux.

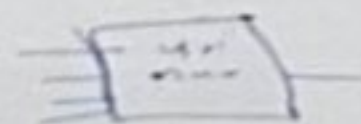
Solution

S_1	S_0	F
0	0	0
0	1	1
1	0	1
1	1	0



b) Implement a full adder circuit using 4:1 multiplexers

[4 marks]



1.

a) $F = P \oplus Q$

P	Q	F
0	0	0
0	1	1
1	0	1
1	1	0

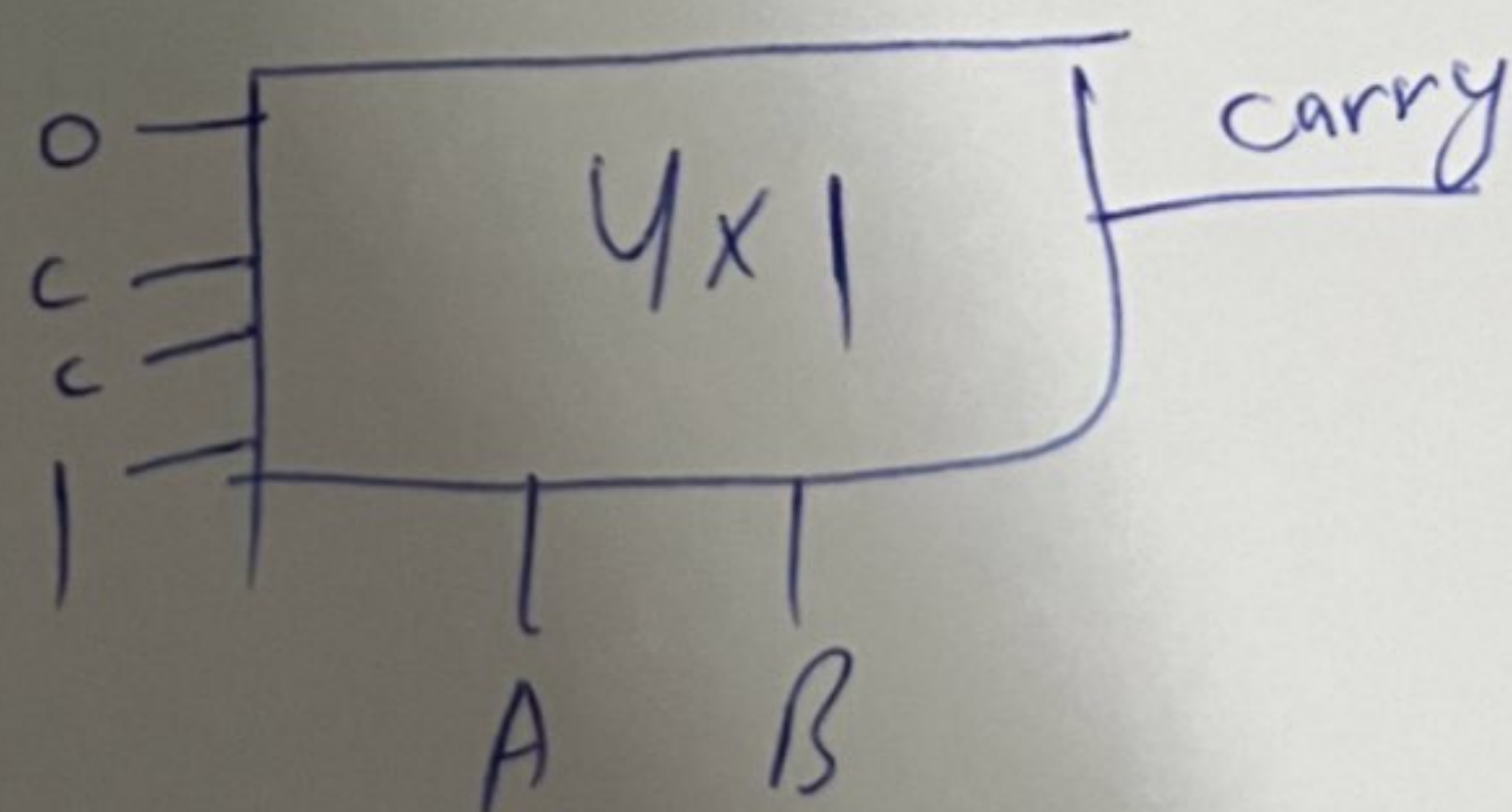
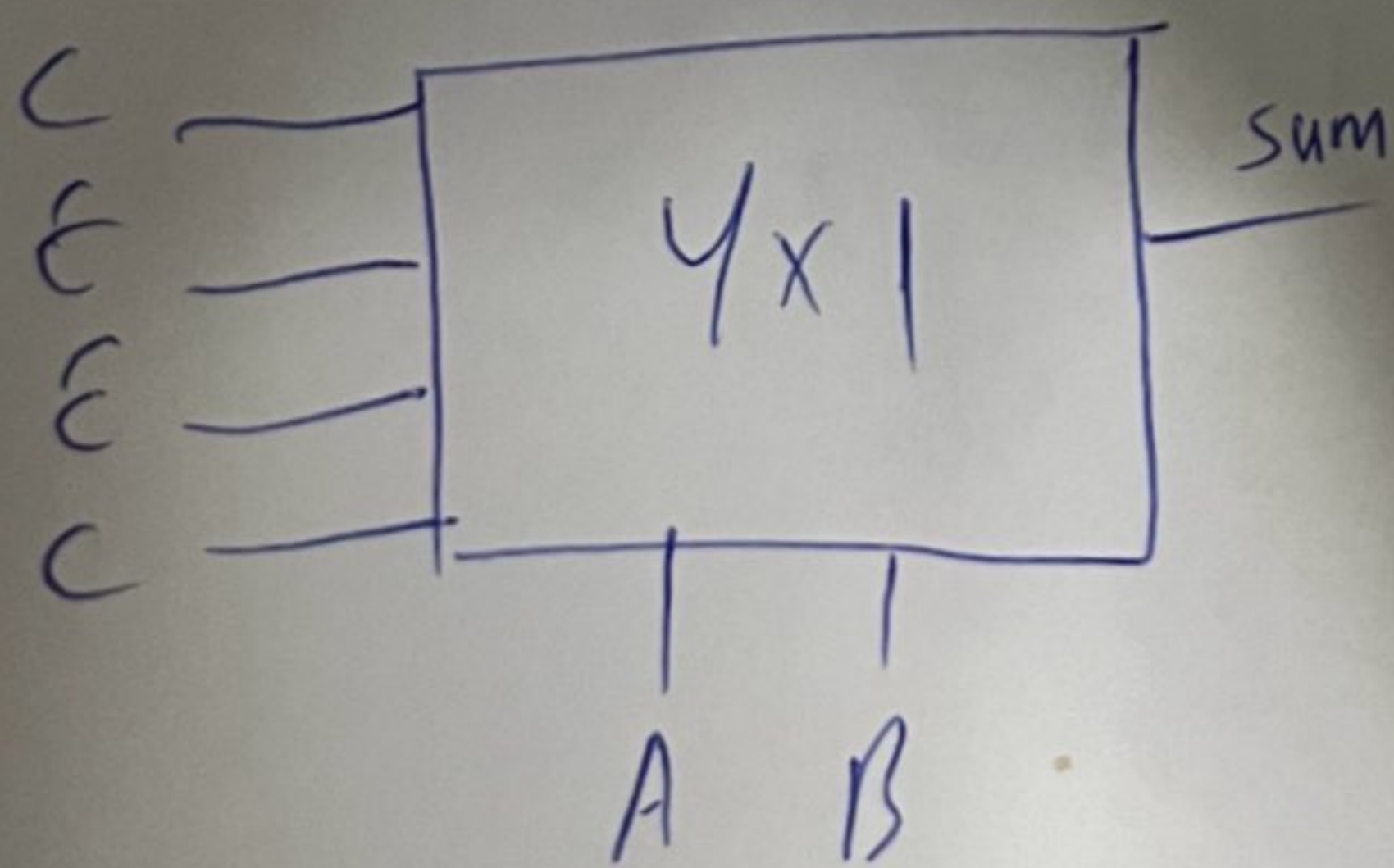
b) لا يفرق

c)

A	B	C	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

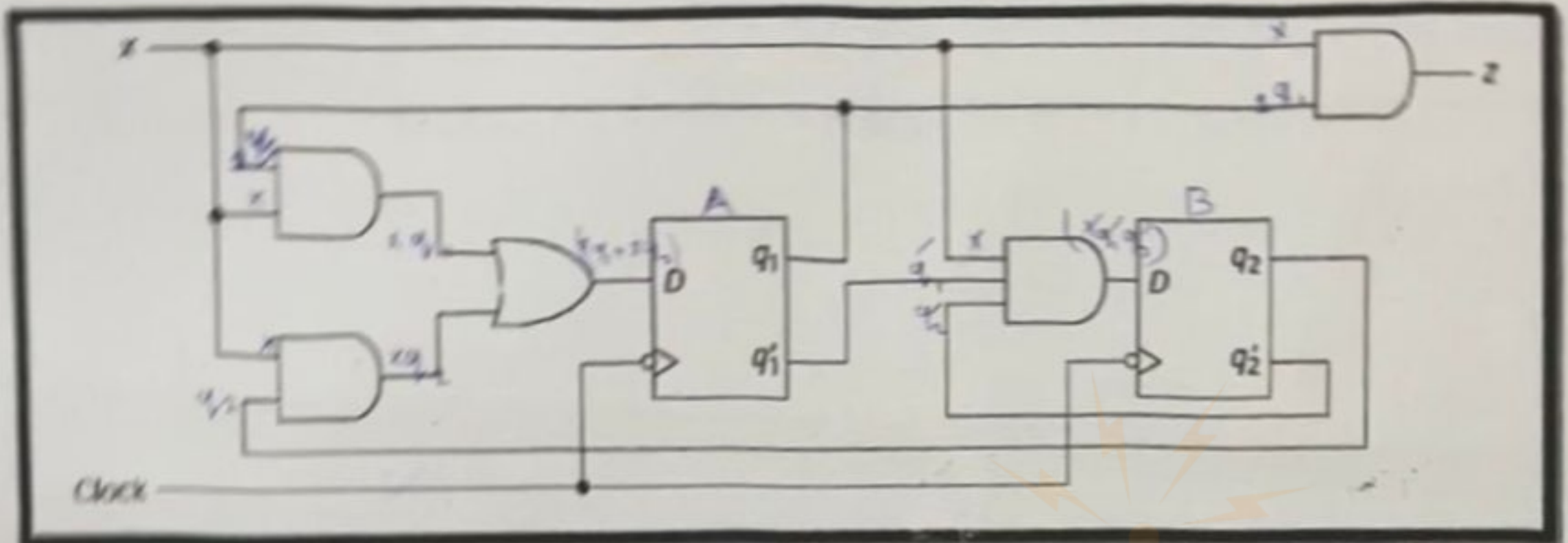
Sum = $\{1, 2, 4, 7\}$
Carry = $\{3, 5, 6, 7\}$

using 4x1 max.



22) Consider the sequential circuit below, the circuit has one input x and one output z , for this circuit determine:

- A) The Next-state and Output equations. [4 marks]
- B) Next-state table [4 marks]
- C) State machine diagram [4 marks]



Solution

$$A(t+1) = xq_1 + xq_2$$

$$= xA + xB$$

$$B(t+1) = xq_1'q_2'$$

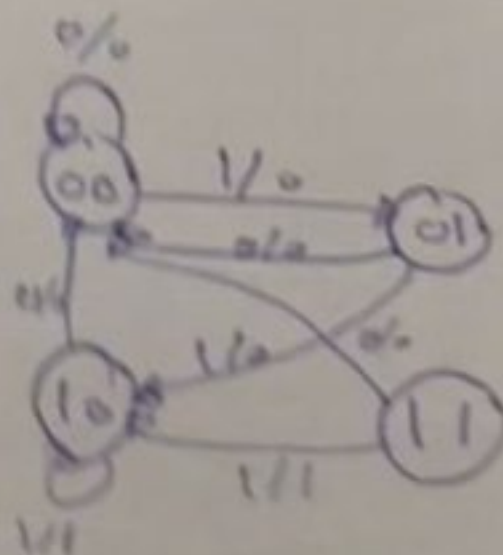
$$= xA'B'$$

$$Z = xq_1$$

$$= xA$$

present	A	B	x	next A	next B	z
0	0	0	0	0	0	0
0	0	0	1	0	1	0
0	0	1	0	0	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	0
0	1	0	1	1	0	1
0	1	1	0	0	0	0
0	1	1	1	1	0	1

$q_1' = D$



Q2

الف

q_1	q_2	X	q_1^+	q_2^+	z
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	0	1
1	1	0	0	0	0
1	1	1	0	1	

$$Q^+ = P \quad \#$$

$$\# \quad q_1^+ = q_1 X + q_2 X$$

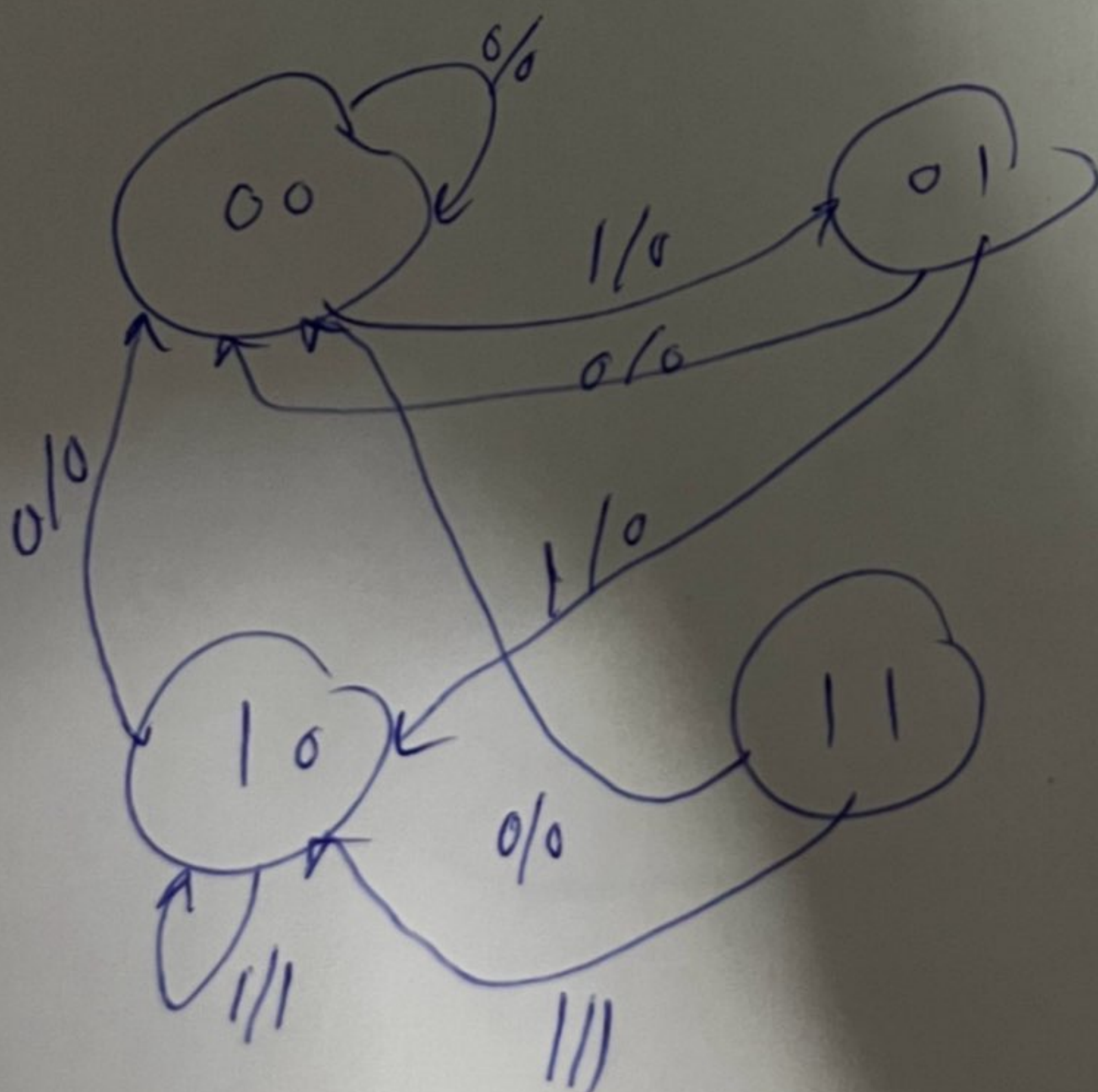
$$\# \quad q_2^+ = X q_1' + q_2'$$

$$\# \quad z = X q_1$$

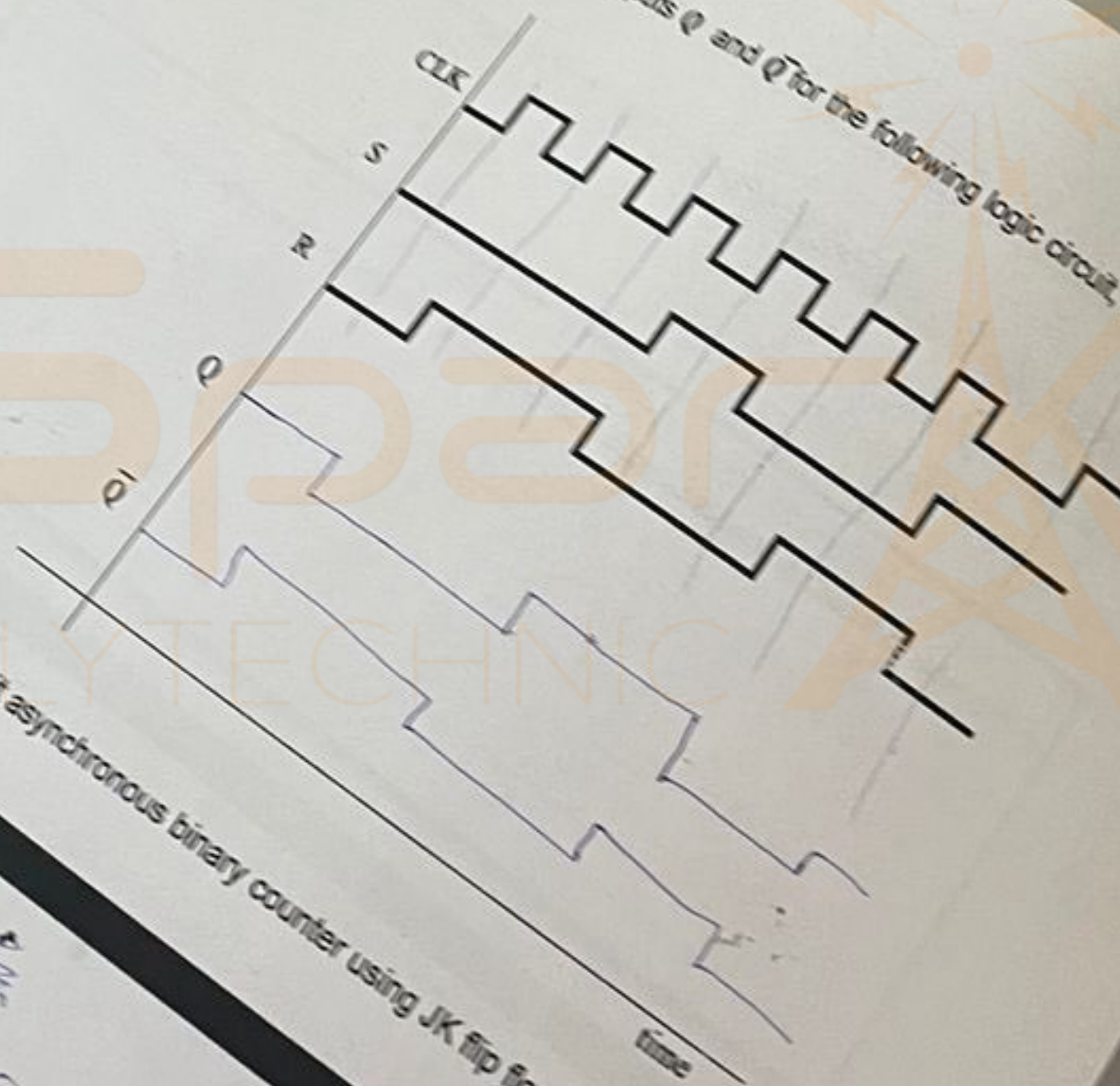


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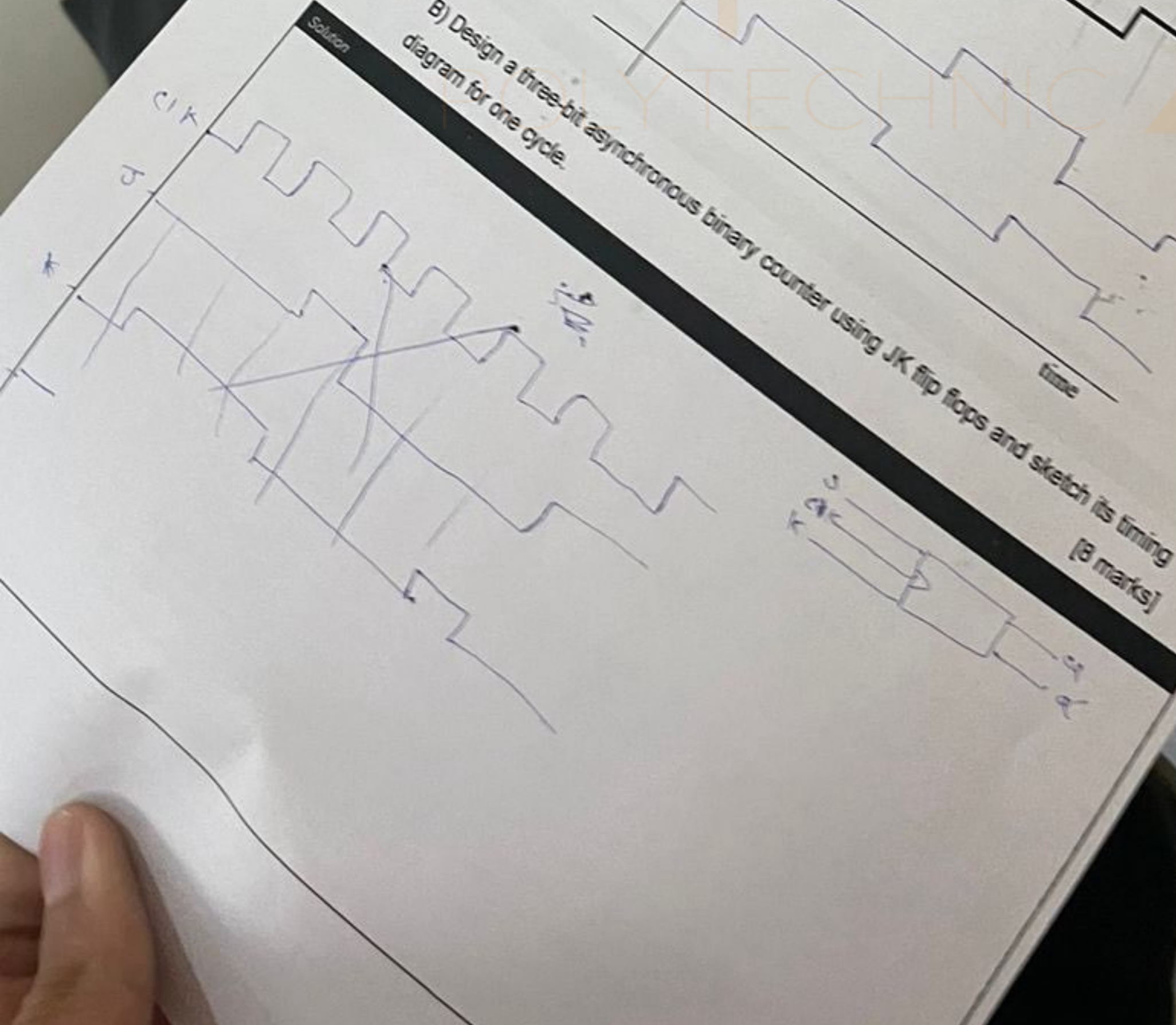
3



Q3) A) Draw the waveforms at outputs Q and $\bar{Q}$ for the following logic circuit, assume the initial value of Q is logic 1.



B) Design a three-bit asynchronous binary counter using JK flip flops and sketch its timing diagram for one cycle. [8 marks]

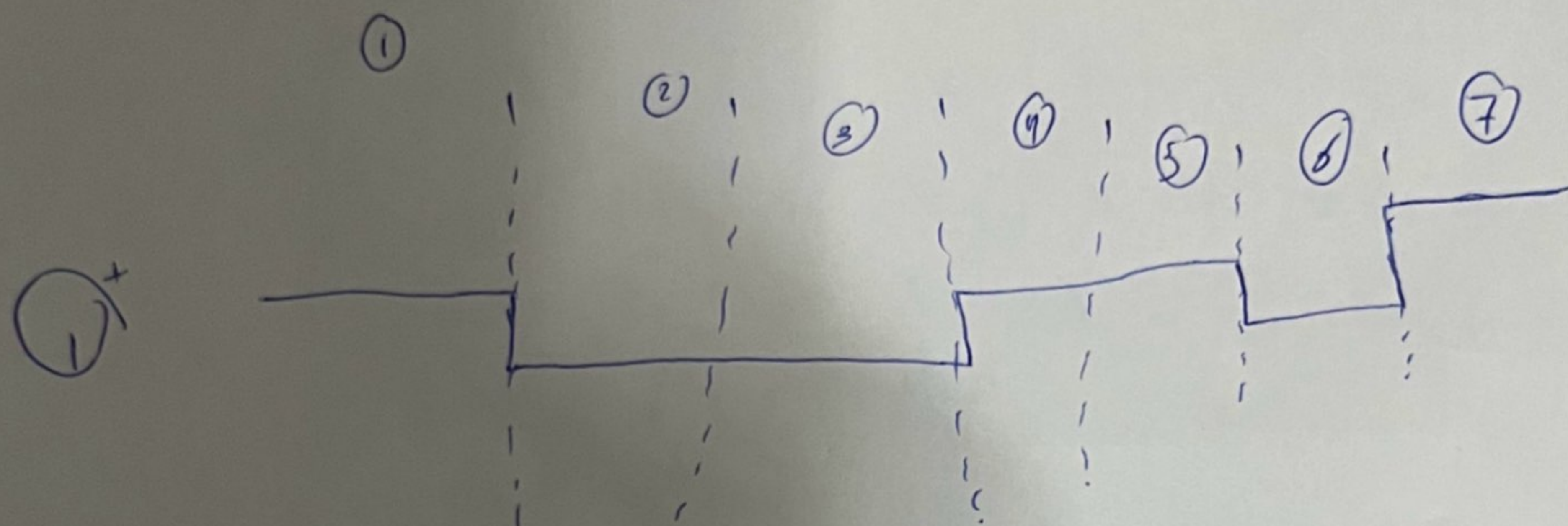


Q4) A) Consider the state table below, draw the state transition diagram.

Q3:

Flip Flop $\rightarrow$ Positive

Q $\rightarrow$ بایست منواله



الغرض ۱۹۸۱

Q3

المسألة الثانية

A B C	A ⁺ B ⁺ C ⁺	J _A K _A	J _B K _B	J _C K _C
0 0 0	0 0 1	0 X	0 X	1 X
0 0 1	0 1 0	0 X	1 X	X 1
0 1 0	0 1 1	0 X	X 0	1 X
0 1 1	1 0 0	1 X	X 1	X 1
1 0 0	1 0 1	X 0	0 X	1 X
1 0 1	1 1 0	X 0	1 X	X 1
1 1 0	1 1 1	X 0	X 0	1 X
1 1 1	0 0 0	X 1	X 1	X 1

Q _A Q _A ⁺	J _A	K _A
0 0	0	X
0 1	1	X
1 0	X	1
1 1	X	0

✓ # $J_A = B C$

✓ # $K_A = B C$

0	0	1	0
1	X	X	X

X	X	X	X
0	0	1	0

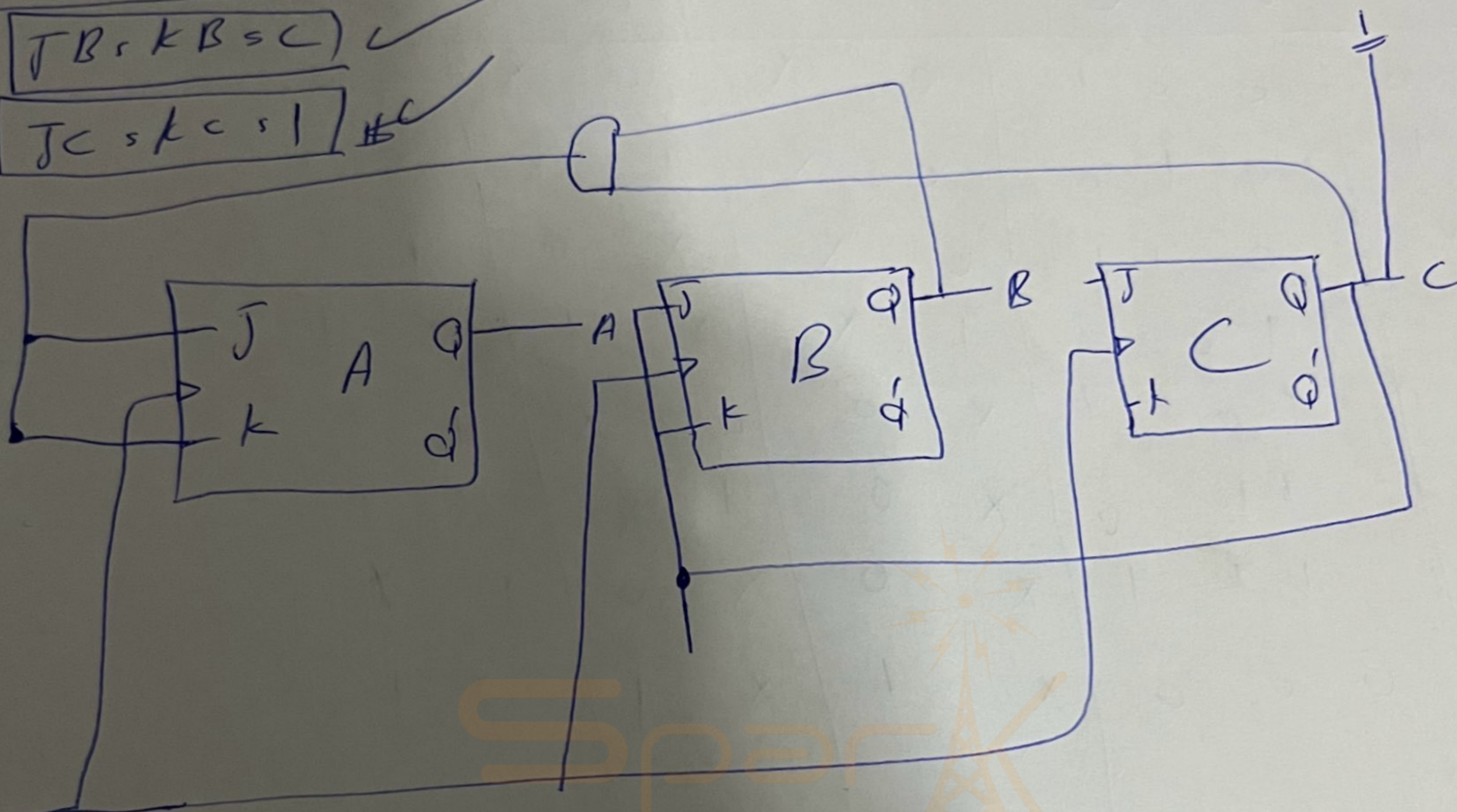
$J_C = J_K = 1$ / B

~~_____~~

0	1	x	x
0	1	x	x

$$JB, KB \subseteq C)$$

$Jc \leq k c \leq 1$



clk

Q4. Reduce the state table below, draw the state diagram corresponding to the reduced state table.

[5 marks]

Present State	Next State		Output	
	x=0	x=1	x=0	x=1
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	e	a	1	0
e	d	c	0	0
f	f	b	1	1
g	e	b	0	1
h	e	a	1	0

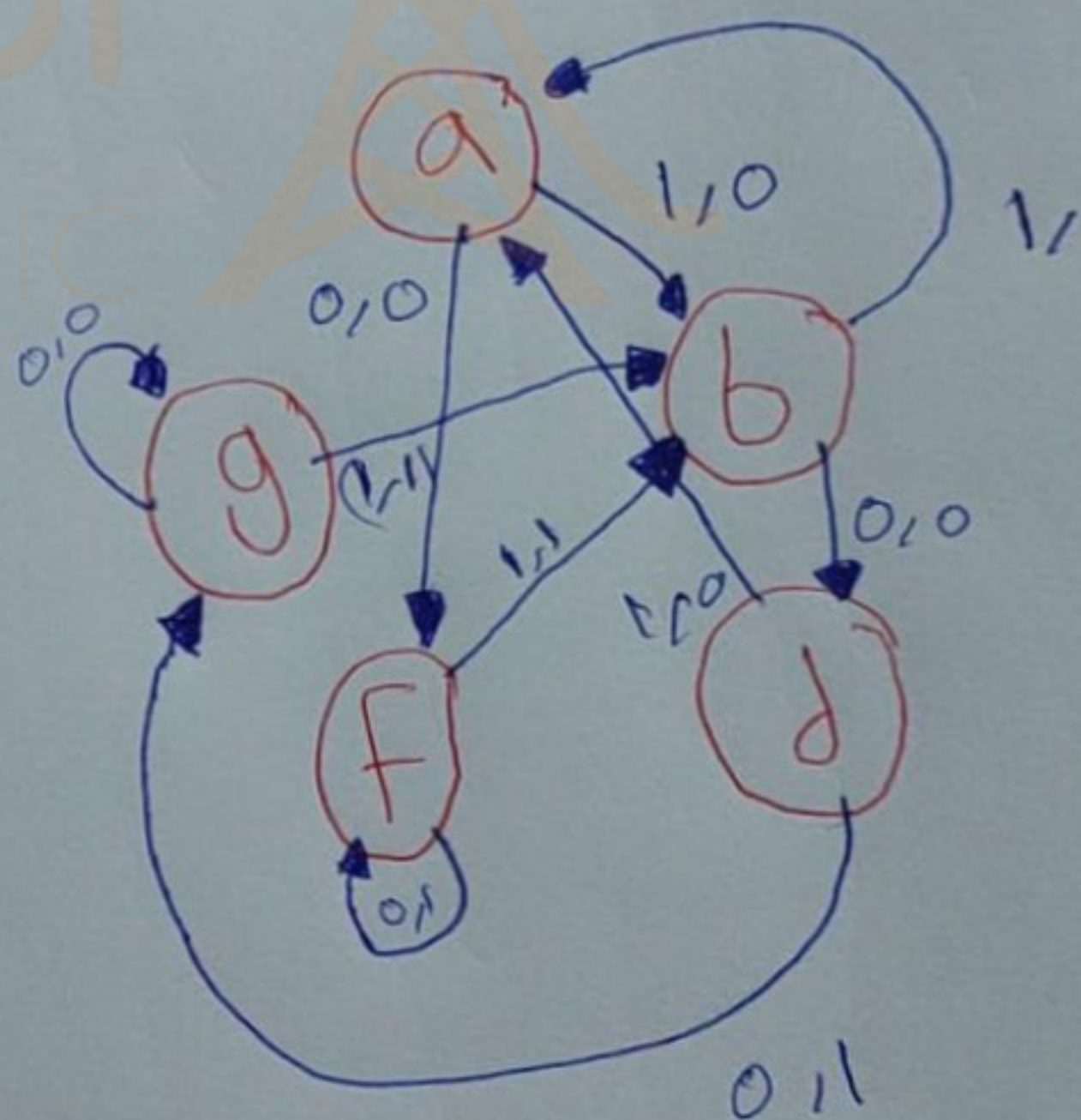
	x=0	x=1	out par x=0	out par x=1
a	f	b	0	0
b	d	c a	0	0
c	f	e b	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	b	0	1
h	g	c a	1	0

c = a (2)

e = b (1)

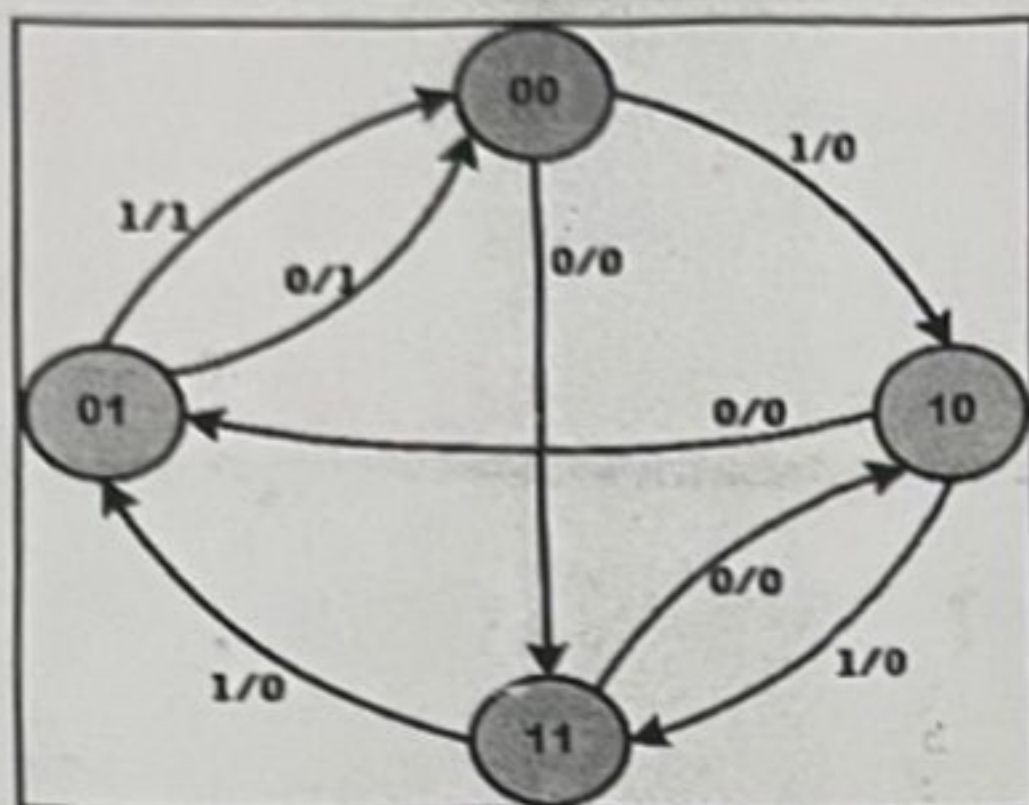
h = d (3)

	x	x	
	0	1	
a	f	b	0 0
b	d	a	0 0
d	g	a	1 0
f	f	b	1 1
g	g	b	0 1



B) Given the following state diagram, design a clocked sequential circuit using JK flip-flops

[8 marks]



Solution

A	B	x	A(t+1)	B(t+1)	y	J _A	K _A	J _B	K _B
0	0	0	1	1	0	1	x	1	x
0	0	1	1	0	0	1	x	0	x
0	1	0	0	0	1	0	x	x	1
0	1	1	0	0	1	0	x	x	1
1	0	0	0	1	0	x	1	1	x
1	0	1	1	1	0	x	0	1	x
1	1	0	1	0	0	x	0	x	1
1	1	1	0	1	0	x	1	x	0

$$J_A = B'$$

B

1	1		
x	x	x	x

$$K_B = A' + x$$

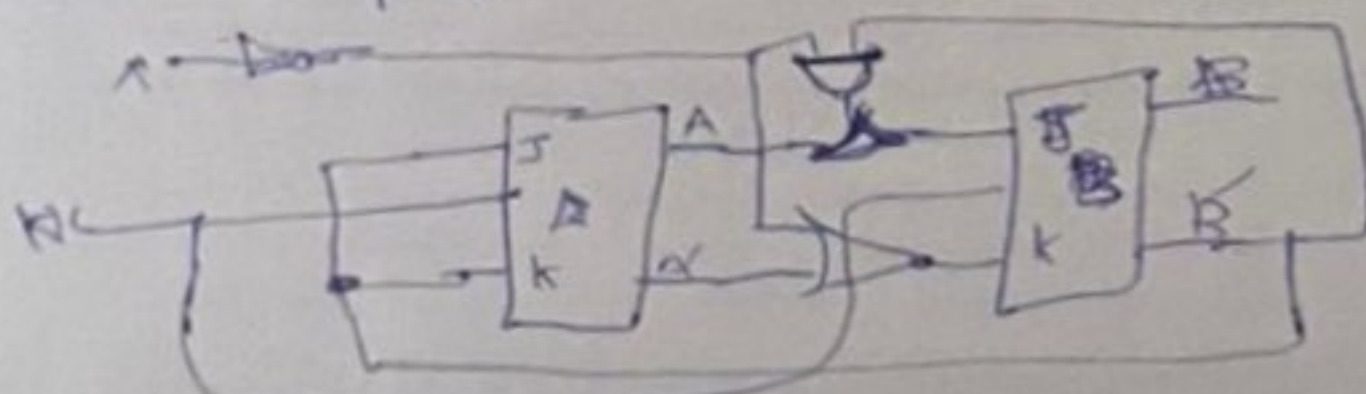
x	x	1	
x	x		

$$K_A = B'$$

x	x	x	x
1	1		

$$J_B = A + x'B$$

1		x	x
1	1	x	x



A	B	X	A ⁺	B ⁺	y	J _A	K _A	J _B	K _B
0	0	0	1	1	0	1	x	1	x
0	0	1	1	0	0	1	x	0	x
0	1	0	0	0	1	0	x	x	1
0	1	1	0	0	1	0	x	x	1
1	0	0	0	1	0	x	1	1	x
1	0	1	1	1	0	x	0	1	x
1	1	0	1	0	0	x	0	x	1
1	1	1	0	1	0	x	1	x	0

Q	Q ⁺	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

$$J_A \begin{array}{|c|c|c|c|} \hline 1 & 1 & & \\ \hline x & x & x & x \\ \hline \end{array} = B'$$

$$K_A \begin{array}{|c|c|c|c|} \hline x & x & x & x \\ \hline 1 & & 1 & \\ \hline \end{array} = BX + B'X' \quad \boxed{B \odot X}$$

$$J_B \begin{array}{|c|c|c|c|} \hline 1 & & x & x \\ \hline 1 & 1 & x & x \\ \hline \end{array} = A + X'$$

$$K_B \begin{array}{|c|c|c|c|} \hline x & x & 1 & 1 \\ \hline x & x & & 1 \\ \hline \end{array} = X' + A$$

$$y \begin{array}{|c|c|c|c|} \hline & & 1 & 1 \\ \hline & & & \\ \hline \end{array} = A'B$$

